05 - zz Routing Basics - EXCERSISE SOLUTIONs

Lesson 3 Exercises: Routing Basics & Static Routes

Complete these exercises to build your routing and troubleshooting skills.

Exercise 1: Deploy, Configure, and Verify End-to-End

Objective: Deploy the hub-and-spoke topology, apply Ansible configuration, and verify full connectivity.

Steps

1. Deploy the topology:

```
cd lessons/clab/03-routing-basics
containerlab deploy -t topology/lab.clab.yml
```

- [x] done

2. Wait for SR Linux to initialize (about 30 seconds), then apply config:

```
cd ansible
ansible-playbook -i inventory.yml playbook.yml
```

- [x] done

3. Verify host-to-router connectivity (same subnet):

```
docker exec clab-routing-basics-host1 ping -c 3 10.1.1.1
docker exec clab-routing-basics-host2 ping -c 3 10.1.4.1
docker exec clab-routing-basics-host3 ping -c 3 10.1.5.1
```

4. Verify end-to-end cross-subnet connectivity (this is the lesson 02 cliffhanger payoff):

```
docker exec clab-routing-basics-host1 ping -c 3 10.1.4.2
docker exec clab-routing-basics-host1 ping -c 3 10.1.5.2
docker exec clab-routing-basics-host2 ping -c 3 10.1.5.2
```

```
[*]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└──>  docker exec clab-routing-basics-host1 ping -c 3 10.1.1.1
PING 10.1.1.1 (10.1.1.1) 56(84) bytes of data.
64 bytes from 10.1.1.1: icmp_seq=1 ttl=64 time=13.7 ms
64 bytes from 10.1.1.1: icmp_seq=2 ttl=64 time=1.59 ms
64 bytes from 10.1.1.1: icmp_seq=3 ttl=64 time=0.793 ms

--- 10.1.1.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 0.793/5.351/13.675/5.894 ms

[*]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└──> docker exec clab-routing-basics-host2 ping -c 3 10.1.4.1
PING 10.1.4.1 (10.1.4.1) 56(84) bytes of data.
64 bytes from 10.1.4.1: icmp_seq=1 ttl=64 time=1.49 ms
64 bytes from 10.1.4.1: icmp_seq=2 ttl=64 time=1.19 ms
64 bytes from 10.1.4.1: icmp_seq=3 ttl=64 time=0.870 ms

--- 10.1.4.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 0.870/1.184/1.489/0.252 ms

[*]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└──>  docker exec clab-routing-basics-host3 ping -c 3 10.1.5.1
PING 10.1.5.1 (10.1.5.1) 56(84) bytes of data.
64 bytes from 10.1.5.1: icmp_seq=1 ttl=64 time=2.03 ms
64 bytes from 10.1.5.1: icmp_seq=2 ttl=64 time=1.84 ms
64 bytes from 10.1.5.1: icmp_seq=3 ttl=64 time=1.55 ms

--- 10.1.5.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2004ms
rtt min/avg/max/mdev = 1.548/1.804/2.028/0.197 ms
```

✅ done

5. Check the routing table on srl1:

```
docker exec -it clab-routing-basics-srl1 sr_cli -c "show network-instance
default route-table ipv4-unicast summary"
```

```
[*]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└──>    docker exec -it clab-routing-basics-srl1 sr_cli -c "show network-
instance default route-table ipv4-unicast summary"
-------------------------------------------------------------------------
-------------------------------------------------------------
IPv4 unicast route table of network instance default
-------------------------------------------------------------------------
-------------------------------------------------------------
+----------+----------+----------+----------+----------+----------+---------
-+----------+----------+----------+----------+----------+
|  Prefix  |    ID    |  Route   |  Route   |  Active  |  Origin  |  Metric
|   Pref   | Next-hop | Next-hop |  Backup  |  Backup  |
|          |          |   Type   |  Owner   |          | Network  |
|          |  (Type)  | Interfac | Next-hop | Next-hop |
|          |          |          |          |          | Instance |
|          |          |    e     |  (Type)  | Interfac |
|          |          |          |          |          |          |
|          |          |          |          |    e     |
+==========+==========+==========+==========+==========+==========+=========
=+==========+==========+==========+==========+==========+
| 10.1.1.0 | 2        | local    | net_inst | True     | default  | 0
| 0        | 10.1.1.1 | ethernet |          |          |
| /24      |          |          | _mgr     |          |          |
|          | (direct) | -1/1.0   |          |          |
| 10.1.1.1 | 2        | host     | net_inst | True     | default  | 0
| 0        | None (ex | None     |          |          |
| /32      |          |          | _mgr     |          |          |
|          | tract)   |          |          |          |
| 10.1.1.2 | 2        | host     | net_inst | True     | default  | 0
| 0        | None (br |          |          |          |
| 55/32    |          |          | _mgr     |          |          |
|          | oadcast) |          |          |          |
| 10.1.2.0 | 3        | local    | net_inst | True     | default  | 0
| 0        | 10.1.2.1 | ethernet |          |          |
| /24      |          |          | _mgr     |          |          |
|          | (direct) | -1/2.0   |          |          |
| 10.1.2.1 | 3        | host     | net_inst | True     | default  | 0
| 0        | None (ex | None     |          |          |
| /32      |          |          | _mgr     |          |          |
|          | tract)   |          |          |          |
| 10.1.2.2 | 3        | host     | net_inst | True     | default  | 0
| 0        | None (br |          |          |          |
| 55/32    |          |          | _mgr     |          |          |
|          | oadcast) |          |          |          |
| 10.1.3.0 | 4        | local    | net_inst | True     | default  | 0
| 0        | 10.1.3.1 | ethernet |          |          |
```

```
| /24      |          |          | _mgr     |          |          |
|          | (direct) | -1/3.0   |          |          |          |
| 10.1.3.1 | 4        | host     | net_inst | True     | default  | 0
| 0        | None (ex | None     |          |          |          |
| /32      |          |          | _mgr     |          |          |
|          | tract)   |          |          |          |          |
| 10.1.3.2 | 4        | host     | net_inst | True     | default  | 0
| 0        | None (br |          |          |          |          |
| 55/32    |          |          | _mgr     |          |          |
|          | oadcast) |          |          |          |          |
| 10.1.4.0 | 0        | static   | static_r | True     | default  | 1
| 5        | 10.1.2.0 | ethernet |          |          |          |
| /24      |          |          | oute_mgr |          |          |
|          | /24 (ind | -1/2.0   |          |          |          |
|          |          |          |          |          |          |
|          | irect/lo |          |          |          |          |
|          |          |          |          |          |          |
|          | cal)     |          |          |          |          |
| 10.1.5.0 | 0        | static   | static_r | True     | default  | 1
| 5        | 10.1.3.0 | ethernet |          |          |          |
| /24      |          |          | oute_mgr |          |          |
|          | /24 (ind | -1/3.0   |          |          |          |
|          |          |          |          |          |          |
|          | irect/lo |          |          |          |          |
|          |          |          |          |          |          |
|          | cal)     |          |          |          |          |
+----------+----------+----------+----------+----------+----------+---------
-+----------+----------+----------+----------+----------+
----------------------------------------------------------------------------
--------------------------------------------------------------
IPv4 routes total                    : 11
IPv4 prefixes with active routes     : 11
IPv4 prefixes with active ECMP routes: 0
----------------------------------------------------------------------------
--------------------------------------------------------------
Parsing error: Unknown token 'commit'. Options are ['!', '#', '/', 'acl',
'back', 'bash', 'bfd', 'cls', 'date', 'diff', 'echo', 'enter', '
environment', 'exit', 'file', 'help', 'history', 'info', 'interface',
'list', 'monitor', 'network-instance', 'passwd', 'ping', 'ping6', 'p
latform', 'pwc', 'qos', 'quit', 'references', 'routing-policy', 'save',
'show', 'source', 'ssh', 'system', 'tcptraceroute', 'tech-support'
, 'tools', 'traceroute', 'traceroute6', 'tree', 'tunnel', 'tunnel-
interface', 'watch', '{']
```

done

Deliverables

- Which pings succeeded (all should work now -- unlike lesson 02)
  - [x] done
- Routing table output from srl1 showing both directly connected and static routes
  - [x] done

Exercise 2: Read the Routing Table

Objective: Understand routing table entries and trace a packet's path hop by hop.

- [x] done

Steps

1. Examine the routing table on all 3 routers:

```
docker exec -it clab-routing-basics-srl1 sr_cli -c "show network-instance default route-table ipv4-unicast summary"
docker exec -it clab-routing-basics-srl2 sr_cli -c "show network-instance default route-table ipv4-unicast summary"
docker exec -it clab-routing-basics-srl3 sr_cli -c "show network-instance default route-table ipv4-unicast summary"
```

- [x] done

srl1

```
[x]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└──>    docker exec -it clab-routing-basics-srl1 sr_cli -c "show network-
instance default route-table ipv4-unicast summary"
---------------------------------------------------------------------------
------------------------------------------------------------
IPv4 unicast route table of network instance default
---------------------------------------------------------------------------
------------------------------------------------------------
+----------+----------+----------+----------+----------+----------+---------
-+----------+----------+----------+----------+----------+
|  Prefix  |    ID    |  Route   |  Route   |  Active  |  Origin  |  Metric
|   Pref   | Next-hop | Next-hop |  Backup  |  Backup  |
|          |          |   Type   |  Owner   |          | Network  |
|          |  (Type)  | Interfac | Next-hop | Next-hop |
|          |          |          |          |          | Instance |
|          |          |    e     |  (Type)  | Interfac |
|          |          |          |          |          |          |
|          |          |          |          |    e     |
+==========+==========+==========+==========+==========+==========+=========
=+==========+==========+==========+==========+==========+
| 10.1.1.0 | 2        | local    | net_inst | True     | default  | 0
| 0        | 10.1.1.1 | ethernet |          |          |
| /24      |          |          | _mgr     |          |          |
|          | (direct) | -1/1.0   |          |          |
| 10.1.1.1 | 2        | host     | net_inst | True     | default  | 0
| 0        | None (ex | None     |          |          |
| /32      |          |          | _mgr     |          |          |
|          | tract)   |          |          |          |
| 10.1.1.2 | 2        | host     | net_inst | True     | default  | 0
| 0        | None (br |          |          |          |
| 55/32    |          |          | _mgr     |          |          |
|          | oadcast) |          |          |          |
| 10.1.2.0 | 3        | local    | net_inst | True     | default  | 0
| 0        | 10.1.2.1 | ethernet |          |          |
| /24      |          |          | _mgr     |          |          |
|          | (direct) | -1/2.0   |          |          |
| 10.1.2.1 | 3        | host     | net_inst | True     | default  | 0
| 0        | None (ex | None     |          |          |
| /32      |          |          | _mgr     |          |          |
|          | tract)   |          |          |          |
| 10.1.2.2 | 3        | host     | net_inst | True     | default  | 0
| 0        | None (br |          |          |          |
| 55/32    |          |          | _mgr     |          |          |
|          | oadcast) |          |          |          |
```

```
| 10.1.3.0 | 4        | local    | net_inst | True     | default  | 0
| 0        | 10.1.3.1 | ethernet |          |          |
| /24      |          |          | _mgr     |          |          |
|          | (direct) | -1/3.0   |          |          |
| 10.1.3.1 | 4        | host     | net_inst | True     | default  | 0
| 0        | None (ex | None     |          |          |
| /32      |          |          | _mgr     |          |          |
|          | tract)   |          |          |          |
| 10.1.3.2 | 4        | host     | net_inst | True     | default  | 0
| 0        | None (br |          |          |          |
| 55/32    |          |          | _mgr     |          |          |
|          | oadcast) |          |          |          |
| 10.1.4.0 | 0        | static   | static_r | True     | default  | 1
| 5        | 10.1.2.0 | ethernet |          |          |
| /24      |          |          | oute_mgr |          |          |
|          | /24 (ind | -1/2.0   |          |          |
|          |          |          |          |          |          |
|          | irect/lo |          |          |          |
|          |          |          |          |          |          |
|          | cal)     |          |          |          |
| 10.1.5.0 | 0        | static   | static_r | True     | default  | 1
| 5        | 10.1.3.0 | ethernet |          |          |
| /24      |          |          | oute_mgr |          |          |
|          | /24 (ind | -1/3.0   |          |          |
|          |          |          |          |          |          |
|          | irect/lo |          |          |          |
|          |          |          |          |          |          |
|          | cal)     |          |          |          |
+----------+----------+----------+----------+----------+----------+---------
-+----------+----------+----------+----------+----------+
-----------------------------------------------------------------------------
--------------------------------------------------------------
IPv4 routes total                    : 11
IPv4 prefixes with active routes     : 11
IPv4 prefixes with active ECMP routes: 0
-----------------------------------------------------------------------------
--------------------------------------------------------------
Parsing error: Unknown token 'commit'. Options are ['!', '#', '/', 'acl',
'back', 'bash', 'bfd', 'cls', 'date', 'diff', 'echo', 'enter', '
environment', 'exit', 'file', 'help', 'history', 'info', 'interface',
'list', 'monitor', 'network-instance', 'passwd', 'ping', 'ping6', 'p
latform', 'pwc', 'qos', 'quit', 'references', 'routing-policy', 'save',
'show', 'source', 'ssh', 'system', 'tcptraceroute', 'tech-support'
, 'tools', 'traceroute', 'traceroute6', 'tree', 'tunnel', 'tunnel-
interface', 'watch', '{']
```

srl2

```
[x]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└──>    docker exec -it clab-routing-basics-srl2 sr_cli -c "show network-
instance default route-table ipv4-unicast summary"
----------------------------------------------------------------------------
--------------------------------------------------------------
IPv4 unicast route table of network instance default
----------------------------------------------------------------------------
--------------------------------------------------------------
+----------+----------+----------+----------+----------+----------+---------
-+----------+----------+----------+----------+----------+
|  Prefix  |    ID    |  Route   |  Route   |  Active  |  Origin  |  Metric
|   Pref   | Next-hop | Next-hop |  Backup  |  Backup  |
|          |          |   Type   |  Owner   |          | Network  |
|          |  (Type)  | Interfac | Next-hop | Next-hop |
|          |          |          |          |          | Instance |
|          |          |    e     |  (Type)  | Interfac |
|          |          |          |          |          |          |
|          |          |          |          |    e     |
+==========+==========+==========+==========+==========+==========+=========
=+==========+==========+==========+==========+==========+
| 10.1.1.0 | 0        | static   | static_r | True     | default  | 1
| 5        | 10.1.2.0 | ethernet |          |          |
| /24      |          |          | oute_mgr |          |          |
|          | /24 (ind | -1/1.0   |          |          |
|          |          |          |          |          |          |
|          | irect/lo |          |          |          |
|          |          |          |          |          |          |
|          | cal)     |          |          |          |
| 10.1.2.0 | 2        | local    | net_inst | True     | default  | 0
| 0        | 10.1.2.2 | ethernet |          |          |
| /24      |          |          | _mgr     |          |          |
|          | (direct) | -1/1.0   |          |          |
| 10.1.2.2 | 2        | host     | net_inst | True     | default  | 0
| 0        | None (ex | None     |          |          |
| /32      |          |          | _mgr     |          |          |
|          | tract)   |          |          |          |
| 10.1.2.2 | 2        | host     | net_inst | True     | default  | 0
| 0        | None (br |          |          |          |
| 55/32    |          |          | _mgr     |          |          |
|          | oadcast) |          |          |          |
| 10.1.3.0 | 0        | static   | static_r | True     | default  | 1
| 5        | 10.1.2.0 | ethernet |          |          |
| /24      |          |          | oute_mgr |          |          |
|          | /24 (ind | -1/1.0   |          |          |
```

```
|          |          |          |          |          |          |
|          | irect/lo |          |          |          |
|          |          |          |          |          |          |
|          | cal)     |          |          |          |
| 10.1.4.0 | 3        | local    | net_inst | True     | default  | 0
| 0        | 10.1.4.1 | ethernet |          |          |
| /24      |          |          | _mgr     |          |          |
|          | (direct) | -1/2.0   |          |          |
| 10.1.4.1 | 3        | host     | net_inst | True     | default  | 0
| 0        | None (ex | None     |          |          |
| /32      |          |          | _mgr     |          |          |
|          | tract)   |          |          |          |
| 10.1.4.2 | 3        | host     | net_inst | True     | default  | 0
| 0        | None (br |          |          |          |
| 55/32    |          |          | _mgr     |          |          |
|          | oadcast) |          |          |          |
| 10.1.5.0 | 0        | static   | static_r | True     | default  | 1
| 5        | 10.1.2.0 | ethernet |          |          |
| /24      |          |          | oute_mgr |          |          |
|          | /24 (ind | -1/1.0   |          |          |
|          |          |          |          |          |          |
|          | irect/lo |          |          |          |
|          |          |          |          |          |          |
|          | cal)     |          |          |          |
+----------+----------+----------+----------+----------+----------+---------
-+----------+----------+----------+----------+----------+
----------------------------------------------------------------------------
--------------------------------------------------------------
IPv4 routes total                    : 9
IPv4 prefixes with active routes     : 9
IPv4 prefixes with active ECMP routes: 0
----------------------------------------------------------------------------
--------------------------------------------------------------
Parsing error: Unknown token 'commit'. Options are ['!', '#', '/', 'acl',
'back', 'bash', 'bfd', 'cls', 'date', 'diff', 'echo', 'enter', '
environment', 'exit', 'file', 'help', 'history', 'info', 'interface',
'list', 'monitor', 'network-instance', 'passwd', 'ping', 'ping6', 'p
latform', 'pwc', 'qos', 'quit', 'references', 'routing-policy', 'save',
'show', 'source', 'ssh', 'system', 'tcptraceroute', 'tech-support'
, 'tools', 'traceroute', 'traceroute6', 'tree', 'tunnel', 'tunnel-
interface', 'watch', '{']
```

srl3

```
[x]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└──>    docker exec -it clab-routing-basics-srl3 sr_cli -c "show network-
instance default route-table ipv4-unicast summary"
---------------------------------------------------------------------------
--------------------------------------------------------------
IPv4 unicast route table of network instance default
---------------------------------------------------------------------------
--------------------------------------------------------------
+----------+----------+----------+----------+----------+----------+---------
-+----------+----------+----------+----------+----------+
|  Prefix  |    ID    |  Route   |  Route   |  Active  |  Origin  |  Metric
|   Pref   | Next-hop | Next-hop |  Backup  |  Backup  |
|          |          |   Type   |  Owner   |          | Network  |
|          |  (Type)  | Interfac | Next-hop | Next-hop |
|          |          |          |          |          | Instance |
|          |          |    e     |  (Type)  | Interfac |
|          |          |          |          |          |          |
|          |          |          |          |    e     |
+==========+==========+==========+==========+==========+==========+=========
=+==========+==========+==========+==========+==========+
| 10.1.1.0 | 0        | static   | static_r | True     | default  | 1
| 5        | 10.1.3.0 | ethernet |          |          |
| /24      |          |          | oute_mgr |          |          |
|          | /24 (ind | -1/1.0   |          |          |
|          |          |          |          |          |          |
|          | irect/lo |          |          |          |
|          |          |          |          |          |          |
|          | cal)     |          |          |          |
| 10.1.2.0 | 0        | static   | static_r | True     | default  | 1
| 5        | 10.1.3.0 | ethernet |          |          |
| /24      |          |          | oute_mgr |          |          |
|          | /24 (ind | -1/1.0   |          |          |
|          |          |          |          |          |          |
|          | irect/lo |          |          |          |
|          |          |          |          |          |          |
|          | cal)     |          |          |          |
| 10.1.3.0 | 2        | local    | net_inst | True     | default  | 0
| 0        | 10.1.3.2 | ethernet |          |          |
| /24      |          |          | _mgr     |          |          |
|          | (direct) | -1/1.0   |          |          |
| 10.1.3.2 | 2        | host     | net_inst | True     | default  | 0
| 0        | None (ex | None     |          |          |
| /32      |          |          | _mgr     |          |          |
|          | tract)   |          |          |          |
```

```
| 10.1.3.2 | 2        | host     | net_inst | True     | default  | 0
| 0        | None (br |          |          |          |
| 55/32    |          |          | _mgr     |          |          |
|          | oadcast) |          |          |          |
| 10.1.4.0 | 0        | static   | static_r | True     | default  | 1
| 5        | 10.1.3.0 | ethernet |          |          |
| /24      |          |          | oute_mgr |          |          |
|          | /24 (ind | -1/1.0   |          |          |
|          |          |          |          |          |          |
|          | irect/lo |          |          |          |
|          |          |          |          |          |          |
|          | cal)     |          |          |          |
| 10.1.5.0 | 3        | local    | net_inst | True     | default  | 0
| 0        | 10.1.5.1 | ethernet |          |          |
| /24      |          |          | _mgr     |          |          |
|          | (direct) | -1/2.0   |          |          |
| 10.1.5.1 | 3        | host     | net_inst | True     | default  | 0
| 0        | None (ex | None     |          |          |
| /32      |          |          | _mgr     |          |          |
|          | tract)   |          |          |          |
| 10.1.5.2 | 3        | host     | net_inst | True     | default  | 0
| 0        | None (br |          |          |          |
| 55/32    |          |          | _mgr     |          |          |
|          | oadcast) |          |          |          |
+----------+----------+----------+----------+----------+----------+---------
-+----------+----------+----------+----------+----------+
----------------------------------------------------------------------------
--------------------------------------------------------------
IPv4 routes total                    : 9
IPv4 prefixes with active routes     : 9
IPv4 prefixes with active ECMP routes: 0
----------------------------------------------------------------------------
--------------------------------------------------------------
Parsing error: Unknown token 'commit'. Options are ['!', '#', '/', 'acl',
'back', 'bash', 'bfd', 'cls', 'date', 'diff', 'echo', 'enter', '
environment', 'exit', 'file', 'help', 'history', 'info', 'interface',
'list', 'monitor', 'network-instance', 'passwd', 'ping', 'ping6', 'p
latform', 'pwc', 'qos', 'quit', 'references', 'routing-policy', 'save',
'show', 'source', 'ssh', 'system', 'tcptraceroute', 'tech-support'
, 'tools', 'traceroute', 'traceroute6', 'tree', 'tunnel', 'tunnel-
interface', 'watch', '{']
```

2. For each router, identify:
   - Which routes are "directly connected" (local)
   - Which routes are "static"

- [x] done

srl1

```
[x]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└──> docker exec -it clab-routing-basics-srl1 sr_cli
Using configuration file(s): []
Welcome to the srlinux CLI.
Type 'help' (and press <ENTER>) if you need any help using this.
--{ + running }--[  ]--
A:srl1# show network-instance default route-table ipv4-unicast summary |grep
-i "local\|static\|prefix.*id"
|  Prefix  |    ID    |  Route   |  Route   |  Active  |  Origin  |  Metric
|   Pref   | Next-hop | Next-hop |  Backup  |  Backup  |
| 10.1.1.0 | 2        | local    | net_inst | True     | default  | 0
| 0        | 10.1.1.1 | ethernet |          |          |
| 10.1.2.0 | 3        | local    | net_inst | True     | default  | 0
| 0        | 10.1.2.1 | ethernet |          |          |
| 10.1.3.0 | 4        | local    | net_inst | True     | default  | 0
| 0        | 10.1.3.1 | ethernet |          |          |
| 10.1.4.0 | 0        | static   | static_r | True     | default  | 1
| 5        | 10.1.2.0 | ethernet |          |          |
| 10.1.5.0 | 0        | static   | static_r | True     | default  | 1
| 5        | 10.1.3.0 | ethernet |          |          |
--{ + running }--[  ]--
```

srl2

```
[*]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└──> docker exec -it clab-routing-basics-srl2 sr_cli
Using configuration file(s): []
Welcome to the srlinux CLI.
Type 'help' (and press <ENTER>) if you need any help using this.
--{ + running }--[  ]--
A:srl2# show network-instance default route-table ipv4-unicast summary |grep
-i "local\|static\|prefix.*id"
|  Prefix  |    ID    |  Route   |  Route   |  Active  |  Origin  |  Metric
|   Pref   | Next-hop | Next-hop |  Backup  |  Backup  |
| 10.1.1.0 | 0        | static   | static_r | True     | default  | 1
| 5        | 10.1.2.0 | ethernet |          |          |
| 10.1.2.0 | 2        | local    | net_inst | True     | default  | 0
| 0        | 10.1.2.2 | ethernet |          |          |
| 10.1.3.0 | 0        | static   | static_r | True     | default  | 1
| 5        | 10.1.2.0 | ethernet |          |          |
| 10.1.4.0 | 3        | local    | net_inst | True     | default  | 0
| 0        | 10.1.4.1 | ethernet |          |          |
| 10.1.5.0 | 0        | static   | static_r | True     | default  | 1
| 5        | 10.1.2.0 | ethernet |          |          |
--{ + running }--[  ]--
```

srl3

```
[x]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└──> docker exec -it clab-routing-basics-srl3 sr_cli
Using configuration file(s): []
Welcome to the srlinux CLI.
Type 'help' (and press <ENTER>) if you need any help using this.
--{ + running }--[  ]--
A:srl3# show network-instance default route-table ipv4-unicast summary |grep
-i "local\|static\|prefix.*id"
|  Prefix  |    ID    |  Route   |  Route   |  Active  |  Origin  |  Metric
|   Pref   | Next-hop | Next-hop |  Backup  |  Backup  |
| 10.1.1.0 | 0        | static   | static_r | True     | default  | 1
| 5        | 10.1.3.0 | ethernet |          |          |
| 10.1.2.0 | 0        | static   | static_r | True     | default  | 1
| 5        | 10.1.3.0 | ethernet |          |          |
| 10.1.3.0 | 2        | local    | net_inst | True     | default  | 0
| 0        | 10.1.3.2 | ethernet |          |          |
| 10.1.4.0 | 0        | static   | static_r | True     | default  | 1
| 5        | 10.1.3.0 | ethernet |          |          |
| 10.1.5.0 | 3        | local    | net_inst | True     | default  | 0
| 0        | 10.1.5.1 | ethernet |          |          |
--{ + running }--[  ]--
```

3. Trace the path of a packet from host2 (10.1.4.2) to host3 (10.1.5.2):
    - Step 1: host2 sends to its default gateway (10.1.4.1 = srl2)
    - Step 2: srl2 looks up 10.1.5.0/24 in its routing table -- what does it find?
    - Step 3: The packet arrives at srl1 -- what route does srl1 use for 10.1.5.0/24?
    - Step 4: The packet arrives at srl3 -- how does it reach host3?

```
[x]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└──> docker exec -it clab-routing-basics-host2 sh
/ # traceroute 10.1.5.2
traceroute to 10.1.5.2 (10.1.5.2), 30 hops max, 60 byte packets
1  10.1.4.1 (10.1.4.1)  0.963 ms  1.438 ms  1.430 ms
2  10.1.2.1 (10.1.2.1)  4.741 ms  4.817 ms  4.829 ms
3  10.1.3.2 (10.1.3.2)  5.329 ms  6.065 ms  6.073 ms
4  10.1.5.2 (10.1.5.2)  4.811 ms  4.817 ms  5.745 ms
```

4. Now trace the RETURN path (host3 -> host2):

```
[*]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└──> docker exec -it clab-routing-basics-host3 sh
/ # traceroute -n 10.1.4.2
traceroute to 10.1.4.2 (10.1.4.2), 30 hops max, 60 byte packets
1  10.1.5.1  0.906 ms  0.859 ms  0.849 ms
2  10.1.3.1  0.696 ms  0.704 ms  0.701 ms
3  10.1.2.2  1.753 ms  1.968 ms  2.018 ms
4  10.1.4.2  1.167 ms  1.364 ms  1.465 ms
```

- What route does srl3 use to reach 10.1.4.0/24?

```
[x]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└──> ssh clab-routing-basics-srl3
Warning: Permanently added 'clab-routing-basics-srl3' (ED25519) to the list
of known hosts.
................................................................
:                  Welcome to Nokia SR Linux!                  :
:              Open Network OS for the NetOps era.             :
:                                                              :
:    This is a freely distributed official container image.    :
:                      Use it - Share it                       :
:                                                              :
: Get started: https://learn.srlinux.dev                       :
: Container:   https://go.srlinux.dev/container-image          :
: Docs:        https://doc.srlinux.dev/24-10                   :
: Rel. notes:  https://doc.srlinux.dev/rn24-10-1               :
: YANG:        https://yang.srlinux.dev/v24.10.1               :
: Discord:     https://go.srlinux.dev/discord                  :
: Contact:     https://go.srlinux.dev/contact-sales            :
................................................................

Last login: Fri May 22 03:46:17 2026 from 3fff:172:20:20::1
Using configuration file(s): ['/home/admin/.srlinuxrc']
Welcome to the srlinux CLI.
Type 'help' (and press <ENTER>) if you need any help using this.

--{ + running }--[  ]--
A:srl3# traceroute -n 10.2.4.2
Using network instance mgmt
traceroute to 10.2.4.2 (10.2.4.2), 30 hops max, 60 byte packets
1  172.20.20.1  8.029 ms  7.505 ms  7.496 ms
2  192.168.178.250  8.461 ms  8.464 ms  8.460 ms
3  192.168.4.1  10.447 ms  12.483 ms  10.428 ms
4  * * *
[..]
8  * * *
^CCommand execution aborted : 'traceroute -n 10.2.4.2'

--{ + running }--[  ]--
```

Something strange happens here. Because the router `srl3` does correctly route packages from `host3` to `host2` ?!

```
[x]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└──> ssh clab-routing-basics-srl3
Warning: Permanently added 'clab-routing-basics-srl3' (ED25519) to the list
of known hosts.
................................................................
:                  Welcome to Nokia SR Linux!                  :
:              Open Network OS for the NetOps era.             :
:                                                              :
:    This is a freely distributed official container image.    :
:                      Use it - Share it                       :
:                                                              :
: Get started: https://learn.srlinux.dev                       :
: Container:   https://go.srlinux.dev/container-image          :
: Docs:        https://doc.srlinux.dev/24-10                   :
: Rel. notes:  https://doc.srlinux.dev/rn24-10-1               :
: YANG:        https://yang.srlinux.dev/v24.10.1               :
: Discord:     https://go.srlinux.dev/discord                  :
: Contact:     https://go.srlinux.dev/contact-sales            :
................................................................

Last login: Fri May 22 03:49:27 2026 from 3fff:172:20:20::1
Using configuration file(s): ['/home/admin/.srlinuxrc']
Welcome to the srlinux CLI.
Type 'help' (and press <ENTER>) if you need any help using this.

--{ + running }--[  ]--
A:srl3# show network-instance mgmt route-table ipv4-unicast summary
-----------------------------------------------------------------------------
-------------------------------------------------------------
IPv4 unicast route table of network instance mgmt
-----------------------------------------------------------------------------
-------------------------------------------------------------
+----------+----------+----------+----------+----------+----------+---------
-+----------+----------+----------+----------+----------+
|  Prefix  |    ID    |  Route   |  Route   |  Active  |  Origin  |  Metric
|   Pref   | Next-hop | Next-hop |  Backup  |  Backup  |
|          |          |   Type   |  Owner   |          | Network  |
|          |  (Type)  | Interfac | Next-hop | Next-hop |
|          |          |          |          |          | Instance |
|          |          |    e     |  (Type)  | Interfac |
|          |          |          |          |          |          |
|          |          |          |          |    e     |
+==========+==========+==========+==========+==========+==========+=========
=+==========+==========+==========+==========+==========+
| 0.0.0.0/ | 1        | dhcp     | dhcp_cli | True     | mgmt     | 0
| 5        | 172.20.2 | mgmt0.0  |          |          |
```

```
| 0         |         |         | ent_mgr |         |         |
|           | 0.1     |         |         |         |         |
|           |         |         |         |         |         |
|           | (direct) |        |         |         |         |
| 172.20.2  | 0       | linux   | linux_mg | False  | mgmt    | 0
| 5         | 172.20.2 | mgmt0.0 |        |         |         |
| 0.0/24    |         |         | r       |         |         |
|           | 0.0     |         |         |         |         |
|           |         |         |         |         |         |
|           | (direct) |        |         |         |         |
| 172.20.2  | 1       | local   | net_inst | True   | mgmt    | 0
| 0         | 172.20.2 | mgmt0.0 |        |         |         |
| 0.0/24    |         |         | _mgr    |         |         |
|           | 0.13    |         |         |         |         |
|           |         |         |         |         |         |
|           | (direct) |        |         |         |         |
| 172.20.2  | 1       | host    | net_inst | True   | mgmt    | 0
| 0         | None (ex | None   |         |         |         |
| 0.13/32   |         |         | _mgr    |         |         |
|           | tract)  |         |         |         |         |
| 172.20.2  | 1       | host    | net_inst | True   | mgmt    | 0
| 0         | None (br |        |         |         |         |
| 0.255/32  |         |         | _mgr    |         |         |
|           | oadcast) |        |         |         |         |
+---------+---------+---------+---------+---------+---------+---------
-+---------+---------+---------+---------+---------+
--------------------------------------------------------------------------
------------------------------------------------------------
IPv4 routes total                    : 5
IPv4 prefixes with active routes     : 4
IPv4 prefixes with active ECMP routes: 0
--------------------------------------------------------------------------
------------------------------------------------------------

--{ + running }--[  ]--
```

The problem seems to be, that the connected `host3` does use an other Routing tabel (`default`) than the router self emitted packages via `ping -c 3 10.1.4.2`. These packages seem to use the routing table calles `mgmt`.

Deliverables

- Routing table screenshots from all 3 routers
  - ✅ done

- Written hop-by-hop trace for host2 -> host3 AND the return path
  - ✅ done

Exercise 3: Break/Fix -- Missing Route

Objective: Understand what happens when a router is missing a route and why both directions can break even when only one router is affected.

Setup (break it)

```
docker exec -it clab-routing-basics-srl2 sr_cli
```

Inside SR Linux:

```
enter candidate
delete / network-instance default static-routes route 10.1.5.0/24
delete / network-instance default next-hop-groups group nhg-10-1-5-0-24
commit now
exit
```

```
└──> docker exec -it clab-routing-basics-srl2 sr_cli
Using configuration file(s): []
Welcome to the srlinux CLI.
Type 'help' (and press <ENTER>) if you need any help using this.
--{ + running }--[  ]--
A:srl2# enter candidate
delete / network-instance default static-routes route 10.1.5.0/24
delete / network-instance default next-hop-groups group nhg-10-1-5-0-24
commit now
exit
All changes have been committed. Leaving candidate mode.
--{ + running }--[  ]--
A:srl2#
EOF encountered
```

Symptom

```
# This FAILS -- host2 cannot reach host3
docker exec clab-routing-basics-host2 ping -c 3 -W 5 10.1.5.2

# This ALSO FAILS -- host3 cannot reach host2 (why?)
docker exec clab-routing-basics-host3 ping -c 3 -W 5 10.1.4.2

# But this WORKS -- host1 can still reach host3
docker exec clab-routing-basics-host1 ping -c 3 10.1.5.2
```

```
[*]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└──> docker exec clab-routing-basics-host2 ping -c 3 -W 5 10.1.5.2
PING 10.1.5.2 (10.1.5.2) 56(84) bytes of data.
From 10.1.4.1 icmp_seq=1 Destination Net Unreachable
From 10.1.4.1 icmp_seq=2 Destination Net Unreachable
From 10.1.4.1 icmp_seq=3 Destination Net Unreachable

--- 10.1.5.2 ping statistics ---
3 packets transmitted, 0 received, +3 errors, 100% packet loss, time 2032ms

[x]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└──> docker exec clab-routing-basics-host3 ping -c 3 -W 5 10.1.4.2
PING 10.1.4.2 (10.1.4.2) 56(84) bytes of data.

--- 10.1.4.2 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2044ms

[x]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└──> docker exec clab-routing-basics-host1 ping -c 3 10.1.5.2
PING 10.1.5.2 (10.1.5.2) 56(84) bytes of data.
64 bytes from 10.1.5.2: icmp_seq=1 ttl=62 time=0.590 ms
64 bytes from 10.1.5.2: icmp_seq=2 ttl=62 time=0.353 ms
64 bytes from 10.1.5.2: icmp_seq=3 ttl=62 time=0.258 ms

--- 10.1.5.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2037ms
rtt min/avg/max/mdev = 0.258/0.400/0.590/0.139 ms
```

Your Task

1. Check srl2's routing table -- what's missing?

```
docker exec -it clab-routing-basics-srl2 sr_cli -c "show network-instance default route-table ipv4-unicast summary"
```

```
[*]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└──> ssh clab-routing-basics-srl2
Warning: Permanently added 'clab-routing-basics-srl2' (ED25519) to the list
of known hosts.
................................................................
:                  Welcome to Nokia SR Linux!                  :
:              Open Network OS for the NetOps era.             :
:                                                              :
:    This is a freely distributed official container image.    :
:                      Use it - Share it                       :
:                                                              :
: Get started: https://learn.srlinux.dev                       :
: Container:   https://go.srlinux.dev/container-image          :
: Docs:        https://doc.srlinux.dev/24-10                   :
: Rel. notes:  https://doc.srlinux.dev/rn24-10-1               :
: YANG:        https://yang.srlinux.dev/v24.10.1               :
: Discord:     https://go.srlinux.dev/discord                  :
: Contact:     https://go.srlinux.dev/contact-sales            :
................................................................

Using configuration file(s): ['/home/admin/.srlinuxrc']
Welcome to the srlinux CLI.
Type 'help' (and press <ENTER>) if you need any help using this.

--{ + running }--[  ]--
A:srl2# show network-instance default route-table ipv4-unicast summary
-----------------------------------------------------------------------------
--------------------------------------------------------------
IPv4 unicast route table of network instance default
-----------------------------------------------------------------------------
--------------------------------------------------------------
+----------+----------+----------+----------+----------+----------+---------
-+----------+----------+----------+----------+----------+
|  Prefix  |    ID    |  Route   |  Route   |  Active  |  Origin  |  Metric
|   Pref   | Next-hop | Next-hop |  Backup  |  Backup  |
|          |          |   Type   |  Owner   |          | Network  |
|          |  (Type)  | Interfac | Next-hop | Next-hop |
|          |          |          |          |          | Instance |
|          |          |    e     |  (Type)  | Interfac |
|          |          |          |          |          |          |
|          |          |          |          |    e     |
+==========+==========+==========+==========+==========+==========+=========
=+==========+==========+==========+==========+==========+
| 10.1.1.0 | 0        | static   | static_r | True     | default  | 1
| 5        | 10.1.2.0 | ethernet |          |          |
| /24      |          |          | oute_mgr |          |          |
```

```
|          | /24 (ind | -1/1.0   |          |          |
|          |          |          |          |          |          |
|          | irect/lo |          |          |          |
|          |          |          |          |          |          |
|          | cal)     |          |          |          |
| 10.1.2.0 | 2        | local    | net_inst | True     | default  | 0
| 0        | 10.1.2.2 | ethernet |          |          |
| /24      |          |          | _mgr     |          |          |
|          | (direct) | -1/1.0   |          |          |
| 10.1.2.2 | 2        | host     | net_inst | True     | default  | 0
| 0        | None (ex | None     |          |          |
| /32      |          |          | _mgr     |          |          |
|          | tract)   |          |          |          |
| 10.1.2.2 | 2        | host     | net_inst | True     | default  | 0
| 0        | None (br |          |          |          |
| 55/32    |          |          | _mgr     |          |          |
|          | oadcast) |          |          |          |
| 10.1.3.0 | 0        | static   | static_r | True     | default  | 1
| 5        | 10.1.2.0 | ethernet |          |          |
| /24      |          |          | oute_mgr |          |          |
|          | /24 (ind | -1/1.0   |          |          |
|          |          |          |          |          |          |
|          | irect/lo |          |          |          |
|          |          |          |          |          |          |
|          | cal)     |          |          |          |
| 10.1.4.0 | 3        | local    | net_inst | True     | default  | 0
| 0        | 10.1.4.1 | ethernet |          |          |
| /24      |          |          | _mgr     |          |          |
|          | (direct) | -1/2.0   |          |          |
| 10.1.4.1 | 3        | host     | net_inst | True     | default  | 0
| 0        | None (ex | None     |          |          |
| /32      |          |          | _mgr     |          |          |
|          | tract)   |          |          |          |
| 10.1.4.2 | 3        | host     | net_inst | True     | default  | 0
| 0        | None (br |          |          |          |
| 55/32    |          |          | _mgr     |          |          |
|          | oadcast) |          |          |          |
+----------+----------+----------+----------+----------+----------+---------
-+----------+----------+----------+----------+----------+
----------------------------------------------------------------------------
--------------------------------------------------------------
IPv4 routes total                    : 8
IPv4 prefixes with active routes     : 8
IPv4 prefixes with active ECMP routes: 0
----------------------------------------------------------------------------
--------------------------------------------------------------
```

2. Explain why both host2->host3 AND host3->host2 fail, even though only srl2 is missing a route (hint: trace both the forward path and the return path for each ping).

The router `srl2` is missing the default route to reach `10.1.5.2`

```
[*]—[kc-debian13-test00-Klon]—[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└──> docker exec -it clab-routing-basics-host2 sh
/ # traceroute -n 10.1.5.2
traceroute to 10.1.5.2 (10.1.5.2), 30 hops max, 60 byte packets
1  10.1.4.1  0.555 ms !N  0.531 ms !N  0.489 ms !N
```

The router `srl3` has intact routes, but the answers from `srl2` never come back to `host3` (via `srl3`) because they never are sent back to `srl3`

```
[*]—[kc-debian13-test00-Klon]—[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└──> docker exec -it clab-routing-basics-host3 sh
/ # traceroute -n 10.1.4.2
traceroute to 10.1.4.2 (10.1.4.2), 30 hops max, 60 byte packets
1  10.1.5.1  1.485 ms  1.453 ms  1.447 ms
2  10.1.3.1  1.331 ms  1.344 ms  1.341 ms
3  * * *
[..]
7  * * *
```

3. Fix by re-adding the route:

   ```
   docker exec -it clab-routing-basics-srl2 sr_cli
   ```

   Inside SR Linux:

   ```
   enter candidate
   set / network-instance default next-hop-groups group nhg-10-1-5-0-24
   admin-state enable
   set / network-instance default next-hop-groups group nhg-10-1-5-0-24
   nexthop 1 ip-address 10.1.2.1
   set / network-instance default static-routes route 10.1.5.0/24 admin-
   state enable
   set / network-instance default static-routes route 10.1.5.0/24 next-hop-
   group nhg-10-1-5-0-24
   commit now
   exit
   ```

```
[x]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└──> docker exec -it clab-routing-basics-srl2 sr_cli
Using configuration file(s): []
Welcome to the srlinux CLI.
Type 'help' (and press <ENTER>) if you need any help using this.
--{ + running }--[  ]--
A:srl2# enter candidate
  set / network-instance default next-hop-groups group nhg-10-1-5-0-24
admin-state enable
  set / network-instance default next-hop-groups group nhg-10-1-5-0-24
nexthop 1 ip-address 10.1.2.1
  set / network-instance default static-routes route 10.1.5.0/24 admin-state
enable
  set / network-instance default static-routes route 10.1.5.0/24 next-hop-
group nhg-10-1-5-0-24
  commit now
  exit
All changes have been committed. Leaving candidate mode.
--{ + running }--[  ]--
A:srl2#
EOF encountered
```

4. Verify the fix:

```
docker exec clab-routing-basics-host2 ping -c 3 10.1.5.2
```

```
[*]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└──>  docker exec clab-routing-basics-host2 ping -c 3 10.1.5.2
PING 10.1.5.2 (10.1.5.2) 56(84) bytes of data.
64 bytes from 10.1.5.2: icmp_seq=1 ttl=61 time=0.800 ms
64 bytes from 10.1.5.2: icmp_seq=2 ttl=61 time=0.767 ms
64 bytes from 10.1.5.2: icmp_seq=3 ttl=61 time=0.637 ms

--- 10.1.5.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2049ms
rtt min/avg/max/mdev = 0.637/0.734/0.800/0.070 ms
```

Deliverables

- Explanation of why the missing route breaks both directions (forward path vs return path)
  ✅ done

- The diagnostic commands you used
  - ✅ done

Exercise 4: Break/Fix -- Wrong Next-Hop (Black Hole)

Objective: Understand what happens when a static route points to a non-existent next-hop.

Setup (break it)

```
docker exec -it clab-routing-basics-srl1 sr_cli
```

Inside SR Linux:

```
enter candidate
set / network-instance default next-hop-groups group nhg-10-1-5-0-24 nexthop
1 ip-address 10.1.3.99
commit now
exit
```

```
[*]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└──> docker exec -it clab-routing-basics-srl1 sr_cli
Using configuration file(s): []
Welcome to the srlinux CLI.
Type 'help' (and press <ENTER>) if you need any help using this.
--{ + running }--[  ]--
A:srl1# enter candidate
set / network-instance default next-hop-groups group nhg-10-1-5-0-24 nexthop
1 ip-address 10.1.3.99
commit now
exit
All changes have been committed. Leaving candidate mode.
--{ + running }--[  ]--
```

Symptom

```
# host1 cannot reach host3
docker exec clab-routing-basics-host1 ping -c 3 -W 5 10.1.5.2
```

```
[x]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└──> docker exec clab-routing-basics-host1 ping -c 3 -W 5 10.1.5.2
PING 10.1.5.2 (10.1.5.2) 56(84) bytes of data.

--- 10.1.5.2 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2033ms
```

Your Task

1. Check srl1's routing table -- does the route exist?

   ```
   docker exec -it clab-routing-basics-srl1 sr_cli -c "show network-instance
   default route-table ipv4-unicast summary"
   ```

```
[x]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└──> docker exec -it clab-routing-basics-srl1 sr_cli -c "show network-
instance default route-table ipv4-unicast summary"
------------------------------------------------------------------------
-------------------------------------------------------------
IPv4 unicast route table of network instance default
------------------------------------------------------------------------
-------------------------------------------------------------
+---------+---------+---------+---------+---------+---------+---------
-+---------+---------+---------+---------+---------+
|  Prefix  |    ID    |  Route   |  Route   |  Active  |  Origin  |  Metric
|   Pref   | Next-hop | Next-hop |  Backup  |  Backup  |
|          |          |   Type   |  Owner   |          | Network  |
|          |  (Type)  | Interfac | Next-hop | Next-hop |
|          |          |          |          |          | Instance |
|          |          |    e     |  (Type)  | Interfac |
|          |          |          |          |          |          |
|          |          |          |          |    e     |
+=========+=========+=========+=========+=========+=========+=========
=+=========+=========+=========+=========+=========+
| 10.1.1.0 | 2        | local    | net_inst | True     | default  | 0
| 0        | 10.1.1.1 | ethernet |          |          |
| /24      |          |          | _mgr     |          |          |
|          | (direct) | -1/1.0   |          |          |
| 10.1.1.1 | 2        | host     | net_inst | True     | default  | 0
| 0        | None (ex | None     |          |          |
| /32      |          |          | _mgr     |          |          |
|          | tract)   |          |          |          |
| 10.1.1.2 | 2        | host     | net_inst | True     | default  | 0
| 0        | None (br |          |          |          |
| 55/32    |          |          | _mgr     |          |          |
|          | oadcast) |          |          |          |
| 10.1.2.0 | 3        | local    | net_inst | True     | default  | 0
| 0        | 10.1.2.1 | ethernet |          |          |
| /24      |          |          | _mgr     |          |          |
|          | (direct) | -1/2.0   |          |          |
| 10.1.2.1 | 3        | host     | net_inst | True     | default  | 0
| 0        | None (ex | None     |          |          |
| /32      |          |          | _mgr     |          |          |
|          | tract)   |          |          |          |
| 10.1.2.2 | 3        | host     | net_inst | True     | default  | 0
| 0        | None (br |          |          |          |
| 55/32    |          |          | _mgr     |          |          |
|          | oadcast) |          |          |          |
| 10.1.3.0 | 4        | local    | net_inst | True     | default  | 0
| 0        | 10.1.3.1 | ethernet |          |          |
```

```
| /24       |           |           | _mgr      |           |           |
|           | (direct)  | -1/3.0    |           |           |
| 10.1.3.1  | 4         | host      | net_inst  | True      | default   | 0
| 0         | None (ex  | None      |           |           |
| /32       |           |           | _mgr      |           |           |
|           | tract)    |           |           |           |
| 10.1.3.2  | 4         | host      | net_inst  | True      | default   | 0
| 0         | None (br  |           |           |           |
| 55/32     |           |           | _mgr      |           |           |
|           | oadcast)  |           |           |           |
| 10.1.4.0  | 0         | static    | static_r  | True      | default   | 1
| 5         | 10.1.2.0  | ethernet  |           |           |
| /24       |           |           | oute_mgr  |           |           |
|           | /24 (ind  | -1/2.0    |           |           |
|           |           |           |           |           |           |
|           | irect/lo  |           |           |           |
|           |           |           |           |           |           |
|           | cal)      |           |           |           |
| 10.1.5.0  | 0         | static    | static_r  | True      | default   | 1
| 5         | 10.1.3.0  | ethernet  |           |           |
| /24       |           |           | oute_mgr  |           |           |
|           | /24 (ind  | -1/3.0    |           |           |
|           |           |           |           |           |           |
|           | irect/lo  |           |           |           |
|           |           |           |           |           |           |
|           | cal)      |           |           |           |
+----------+----------+----------+----------+----------+----------+---------
-+----------+----------+----------+----------+----------+
----------------------------------------------------------------------------
--------------------------------------------------------------
IPv4 routes total                    : 11
IPv4 prefixes with active routes     : 11
IPv4 prefixes with active ECMP routes: 0
----------------------------------------------------------------------------
--------------------------------------------------------------
Parsing error: Unknown token 'commit'. Options are ['!', '#', '/', 'acl',
'back', 'bash', 'bfd', 'cls', 'date', 'diff', 'echo', 'enter', '
environment', 'exit', 'file', 'help', 'history', 'info', 'interface',
'list', 'monitor', 'network-instance', 'passwd', 'ping', 'ping6', 'p
latform', 'pwc', 'qos', 'quit', 'references', 'routing-policy', 'save',
'show', 'source', 'ssh', 'system', 'tcptraceroute', 'tech-support'
, 'tools', 'traceroute', 'traceroute6', 'tree', 'tunnel', 'tunnel-
interface', 'watch', '{']
```

Problem: My environment does not reach the broken state that is described in the solution

```
--{ + running }--[  ]--
A:srl1# show network-instance default route-table ipv4-unicast summary
-----------------------------------------------------------------------------------------------------------------------------------------------
IPv4 unicast route table of network instance default
-----------------------------------------------------------------------------------------------------------------------------------------------
+-----------------------------+-------+------------+----------------------+----------+----------+---------+------------+--------------------+--------------------+
|           Prefix            |  ID   | Route Type |     Route Owner      |  Active  |  Origin  | Metric  |    Pref    |  Next-hop (Type)   | Next-hop Interface |
|                             |       |            |                      |          | Network  |         |            |                    |                    |
|                             |       |            |                      |          | Instance |         |            |                    |                    |
+=============================+=======+============+======================+==========+==========+=========+============+====================+====================+
| 10.1.1.0/24                 | 2     | local      | net_inst_mgr         | True     | default  | 0       | 0          | 10.1.1.1 (direct)  | ethernet-1/1.0     |
| 10.1.1.1/32                 | 2     | host       | net_inst_mgr         | True     | default  | 0       | 0          | None (extract)     | None               |
| 10.1.1.255/32               | 2     | host       | net_inst_mgr         | True     | default  | 0       | 0          | None (broadcast)   |                    |
| 10.1.2.0/24                 | 3     | local      | net_inst_mgr         | True     | default  | 0       | 0          | 10.1.2.1 (direct)  | ethernet-1/2.0     |
| 10.1.2.1/32                 | 3     | host       | net_inst_mgr         | True     | default  | 0       | 0          | None (extract)     | None               |
| 10.1.2.255/32               | 3     | host       | net_inst_mgr         | True     | default  | 0       | 0          | None (broadcast)   |                    |
| 10.1.3.0/24                 | 4     | local      | net_inst_mgr         | True     | default  | 0       | 0          | 10.1.3.1 (direct)  | ethernet-1/3.0     |
| 10.1.3.1/32                 | 4     | host       | net_inst_mgr         | True     | default  | 0       | 0          | None (extract)     | None               |
| 10.1.3.255/32               | 4     | host       | net_inst_mgr         | True     | default  | 0       | 0          | None (broadcast)   |                    |
| 10.1.4.0/24                 | 0     | static     | static_route_mgr     | True     | default  | 1       | 5          | 10.1.2.0/24        | ethernet-1/2.0     |
|                             |       |            |                      |          |          |         |            | (indirect/local)   |                    |
| 10.1.5.0/24                 | 0     | static     | static_route_mgr     | True     | default  | 1       | 5          | 10.1.3.0/24        | ethernet-1/3.0     |
|                             |       |            |                      |          |          |         |            | (indirect/local)   |                    |
+-----------------------------+-------+------------+----------------------+----------+----------+---------+------------+--------------------+--------------------+
-----------------------------------------------------------------------------------------------------------------------------------------------
IPv4 routes total                    : 11
IPv4 prefixes with active routes     : 11
IPv4 prefixes with active ECMP routes: 0
-----------------------------------------------------------------------------------------------------------------------------------------------
```

I never get to the point where the next-hop is literally `10.1.3.99` evenb if I executed all commands as stated above

```
**srl1's routing table after the change:**

```
A:srl1# show network-instance default route-table ipv4-unicast summary
+------------+-------+------------+-------------------+----------+
| Prefix | Type | Route-Owner| Next-hop | Metric |
+============+=======+============+===================+==========+
10.1.1.0/24	local	net_inst	ethernet-1/1.0	0
10.1.2.0/24	local	net_inst	ethernet-1/2.0	0
10.1.3.0/24	local	net_inst	ethernet-1/3.0	0
10.1.4.0/24	static	static	10.1.2.2	1
10.1.5.0/24	static	static	10.1.3.99	1
+------------+-------+------------+-------------------+----------+
```
```

2. The route exists but points to 10.1.3.99. Check the ARP table:

```
docker exec -it clab-routing-basics-srl1 sr_cli -c "show arpnd arp-entries"
```

```
[x]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└──> docker exec -it clab-routing-basics-srl1 sr_cli -c "show arpnd arp-
entries"
+-------------+-------------+----------------+-------------+---------------
---------+-----------------------------------------------------+
|  Interface  | Subinterfac |    Neighbor    |   Origin    |   Link layer
address    |                     Expiry                      |
|             |      e      |                |             |
|                                                  |
+=============+=============+================+=============+===============
========+=====================================================+
| ethernet-   |           0 |       10.1.1.2 |     dynamic |
AA:C1:AB:C5:FA:06       | 3 hours from now                                   |
| 1/1         |             |                |             |
|                                                  |
| ethernet-   |           0 |       10.1.2.2 |     dynamic |
1A:33:04:FF:00:01       | 2 hours from now                                   |
| 1/2         |             |                |             |
|                                                  |
| mgmt0       |           0 |    172.20.20.1 |     dynamic |
BA:4B:B5:4E:81:AA       | 2 hours from now                                   |
+-------------+-------------+----------------+-------------+---------------
---------+-----------------------------------------------------+
---------------------------------------------------------------------------
--------------------------------------------------------------
 Total entries : 3 (0 static, 3 dynamic)
---------------------------------------------------------------------------
--------------------------------------------------------------
Parsing error: Unknown token 'commit'. Options are ['!', '#', '/', 'acl',
'back', 'bash', 'bfd', 'cls', 'date', 'diff', 'echo', 'enter', '
environment', 'exit', 'file', 'help', 'history', 'info', 'interface',
'list', 'monitor', 'network-instance', 'passwd', 'ping', 'ping6', 'p
latform', 'pwc', 'qos', 'quit', 'references', 'routing-policy', 'save',
'show', 'source', 'ssh', 'system', 'tcptraceroute', 'tech-support'
, 'tools', 'traceroute', 'traceroute6', 'tree', 'tunnel', 'tunnel-
interface', 'watch', '{']
```

Problem: My environment does not reach the broken state that is described in the solution

```
**ARP table on srl1:**

```
A:srl1# show arpnd arp-entries
+-----------+-----------+-----------+-----------+
| Interface | IP Addr | MAC | State |
+===========+===========+===========+===========+
e1-1.0	10.1.1.2	aa:c1:...	reachable
e1-2.0	10.1.2.2	aa:c1:...	reachable
e1-3.0	10.1.3.2	aa:c1:...	reachable
e1-3.0	10.1.3.99		incomplete
+-----------+-----------+-----------+-----------+
```
```

3. Explain why a valid-looking route can still black-hole traffic.
   - Because a valid looking route can still have an invalid target like `10.1.3.99`
4. Fix by restoring the correct next-hop:

```
docker exec -it clab-routing-basics-srl1 sr_cli
```

Inside SR Linux:

```
enter candidate
set / network-instance default next-hop-groups group nhg-10-1-5-0-24
nexthop 1 ip-address 10.1.3.2
commit now
exit
```

5. Verify:

```
docker exec clab-routing-basics-host1 ping -c 3 10.1.5.2
```

Deliverables

- Explanation of black hole routing
  - If the route a router would need to send packages bag does not exist, and the returning packages go throu the default route (and never return to the originating host)
- ARP output showing failed resolution for 10.1.3.99
  - I cant complete this excersise, because the setup that I should have for my experimentation does not show up after following all instructions

---

Exercise 5: Break/Fix -- Routing Loop

Objective: Observe what happens when two routers send traffic back and forth in a loop, and understand how TTL prevents infinite loops.

Setup (break it)

```
docker exec -it clab-routing-basics-srl1 sr_cli
```

Inside SR Linux:

```
enter candidate
set / network-instance default next-hop-groups group nhg-10-1-5-0-24 nexthop
1 ip-address 10.1.2.2
commit now
exit
```

```
[*]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-kubecraft/lessons/clab/03-routing-basics/ansible]
└──> docker exec -it clab-routing-basics-srl1 sr_cli
Using configuration file(s): []
Welcome to the srlinux CLI.
Type 'help' (and press <ENTER>) if you need any help using this.
--{ + running }--[  ]--
A:srl1# enter candidate
set / network-instance default next-hop-groups group nhg-10-1-5-0-24 nexthop 1 ip-address 10.1.2.2
commit now
exit
All changes have been committed. Leaving candidate mode.
--{ + running }--[  ]--
A:srl1#
```

Now srl1 sends 10.1.5.0/24 traffic to srl2, and srl2 sends it back to srl1 (via its existing route through the hub).

Symptom

```
# Ping fails
docker exec clab-routing-basics-host1 ping -c 3 -W 10 10.1.5.2

# Traceroute reveals the loop
docker exec clab-routing-basics-host1 traceroute -n -w 2 10.1.5.2
```

```
[*]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└──> docker exec clab-routing-basics-host1 ping -c 3 -W 10 10.1.5.2
PING 10.1.5.2 (10.1.5.2) 56(84) bytes of data.
From 10.1.2.1 icmp_seq=1 Redirect Network(New nexthop: 10.1.2.2)
From 10.1.2.1 icmp_seq=1 Redirect Network(New nexthop: 10.1.2.2)
From 10.1.2.1 icmp_seq=1 Redirect Network(New nexthop: 10.1.2.2)

--- 10.1.5.2 ping statistics ---
1 packets transmitted, 0 received, +3 errors, 100% packet loss, time 0ms


[x]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└──> docker exec clab-routing-basics-host1 traceroute -n -w 2 10.1.5.2
traceroute to 10.1.5.2 (10.1.5.2), 30 hops max, 60 byte packets
1  10.1.1.1  1.353 ms  1.336 ms  1.331 ms
2  10.1.2.2  1.416 ms  1.416 ms  1.415 ms
3  10.1.2.1  2.414 ms  2.420 ms  2.417 ms
4  10.1.2.2  1.818 ms  1.911 ms  2.423 ms
5  * * *
6  10.1.2.2  2.533 ms  2.077 ms  2.076 ms
7  * * *
8  10.1.2.2  2.687 ms * *
9  * * *
10  * * *
11  * * *
12  10.1.2.2  12.756 ms  12.767 ms  12.766 ms
13  * * *
14  10.1.2.2  12.735 ms  11.977 ms  11.970 ms
15  * * *
16  10.1.2.2  10.881 ms  10.781 ms  10.626 ms
17  * * *
18  10.1.2.2  1.858 ms * *
19  * * *
20  * 10.1.2.2  4.004 ms  3.850 ms
21  * * *
22  10.1.2.2  4.708 ms  4.709 ms  1.732 ms
23  * * *
24  10.1.2.2  1.898 ms  0.968 ms  0.903 ms
25  * * *
26  10.1.2.2  13.006 ms  13.024 ms *
27  * * *
28  * * *^C
```

Your Task

1. Run traceroute and observe the alternating hops between srl1 and srl2.
2. Explain: What is TTL? Why does the packet eventually stop?
   - The `TTL` is thenumber of maximum hops a package may take, before it is discarded
3. Fix by restoring srl1's correct next-hop:

   ```
   docker exec -it clab-routing-basics-srl1 sr_cli
   ```

   Inside SR Linux:

   ```
   enter candidate
   set / network-instance default next-hop-groups group nhg-10-1-5-0-24
   nexthop 1 ip-address 10.1.3.2
   commit now
   exit
   ```

```
[*]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└──>  docker exec -it clab-routing-basics-srl1 sr_cli
Using configuration file(s): []
Welcome to the srlinux CLI.
Type 'help' (and press <ENTER>) if you need any help using this.
--{ + running }--[  ]--
A:srl1# enter candidate
  set / network-instance default next-hop-groups group nhg-10-1-5-0-24
nexthop 1 ip-address 10.1.3.2
  commit now
  exit

All changes have been committed. Leaving candidate mode.

```

4. Verify:

   ```
   docker exec clab-routing-basics-host1 ping -c 3 10.1.5.2
   ```

```
[x]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└──> docker exec clab-routing-basics-host1 ping -c 3 10.1.5.2
PING 10.1.5.2 (10.1.5.2) 56(84) bytes of data.
64 bytes from 10.1.5.2: icmp_seq=1 ttl=62 time=0.786 ms
64 bytes from 10.1.5.2: icmp_seq=2 ttl=62 time=0.478 ms
64 bytes from 10.1.5.2: icmp_seq=3 ttl=62 time=0.382 ms

--- 10.1.5.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2038ms
rtt min/avg/max/mdev = 0.382/0.548/0.786/0.172 ms
```

Deliverables

- Traceroute output showing the routing loop pattern
  - ✅ done
- Explanation of TTL and how it prevents infinite loops

✅ done

Exercise 6: Break/Fix -- Unreachable Next-Hop (Link Down)

Objective: Understand that a route can exist in the routing table even when the physical path is broken.

Setup (break it)

```
docker exec -it clab-routing-basics-srl1 sr_cli
```

Inside SR Linux:

```
enter candidate
set / interface ethernet-1/3 admin-state disable
commit now
exit
```

Symptom

```
# host1 cannot reach host3
docker exec clab-routing-basics-host1 ping -c 3 -W 5 10.1.5.2
```

```
└──> docker exec -it clab-routing-basics-srl1 sr_cli
Using configuration file(s): []
Welcome to the srlinux CLI.
Type 'help' (and press <ENTER>) if you need any help using this.
--{ + running }--[  ]--
A:srl1# enter candidate
set / interface ethernet-1/3 admin-state disable
commit now
exit
All changes have been committed. Leaving candidate mode.
--{ + running }--[  ]--
A:srl1#
EOF encountered

[*]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└──> docker exec clab-routing-basics-host1 ping -c 3 -W 5 10.1.5.2
PING 10.1.5.2 (10.1.5.2) 56(84) bytes of data.
From 10.1.1.1 icmp_seq=1 Destination Net Unreachable
From 10.1.1.1 icmp_seq=2 Destination Net Unreachable
^C
```

Your Task

1. Check interface status on srl1:

   ```
   docker exec -it clab-routing-basics-srl1 sr_cli -c "show interface brief"
   ```

```
A:srl1# show interface brief
+---------------------+-----------------+-----------------+-----------------
+-----------------+------------------+
|        Port         |   Admin State   |   Oper State    |       Speed
|       Type        |   Description    |
+=====================+=================+=================+=================
+=================+==================+
| ethernet-1/1        | enable          | up              | 25G
|                   |                  |
| ethernet-1/2        | enable          | up              | 25G
|                   |                  |
| ethernet-1/3        | disable         | down            | 25G
|                   |                  |
| ethernet-1/4        | disable         | down            | 25G
|                   |                  |
```

2. Notice ethernet-1/3 is admin-disabled. The route to 10.1.5.0/24 may still exist but the next-hop is unreachable.
3. Explain: Why can a route exist even when the link is down?
   - (From solutions:) *Static routes are manually configured entries. The router does not automatically remove them when a link goes down. Dynamic routing protocols like BGP or OSPF would detect the link failure and withdraw the route from the table, but static routes are "dumb" -- they stay in the table regardless of link state.*
4. Fix by re-enabling the interface:

   ```
   docker exec -it clab-routing-basics-srl1 sr_cli
   ```

   Inside SR Linux:

   ```
   enter candidate
   set / interface ethernet-1/3 admin-state enable
   commit now
   exit
   ```
5. Verify:

   ```
   docker exec clab-routing-basics-host1 ping -c 3 10.1.5.2
   ```

```
[*]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/an
sible]
└──>    docker exec -it clab-routing-basics-srl1 sr_cli
Using configuration file(s): []
Welcome to the srlinux CLI.
Type 'help' (and press <ENTER>) if you need any help using this.
--{ + running }--[  ]--
A:srl1# enter candidate
  set / interface ethernet-1/3 admin-state enable
  commit now
  exit

All changes have been committed. Leaving candidate mode.
--{ + running }--[  ]--
A:srl1#
EOF encountered

[*]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/an
sible]
└──>    docker exec clab-routing-basics-host1 ping -c 3 10.1.5.2
PING 10.1.5.2 (10.1.5.2) 56(84) bytes of data.
64 bytes from 10.1.5.2: icmp_seq=1 ttl=62 time=0.481 ms
64 bytes from 10.1.5.2: icmp_seq=2 ttl=62 time=0.260 ms
64 bytes from 10.1.5.2: icmp_seq=3 ttl=62 time=0.248 ms

--- 10.1.5.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2009ms
rtt min/avg/max/mdev = 0.248/0.329/0.481/0.107 ms
```

Deliverables

- Interface status output showing the disabled link
  - ✅ done
- Explanation of why route existence does not guarantee path availability
  - ✅ done

Cleanup

After completing all exercises:

```
# Destroy the lab
containerlab destroy -t topology/lab.clab.yml --cleanup

# Verify
docker ps | grep clab
```

Validation

Run the automated tests:

```
pytest tests/ -v
```

```
[x]─[kc-debian13-test00-Klon]─[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics]
└──> pytest tests/ -v
============================================== test session starts
==============================================
platform linux -- Python 3.13.5, pytest-8.3.5, pluggy-1.5.0 --
/usr/bin/python3
cachedir: .pytest_cache
rootdir: /kubecraft/github_rbeckesch_DrewElliot-kubecraft
configfile: pyproject.toml
plugins: anyio-4.8.0
collected 22 items

tests/test_routing_basics.py::TestEnvironment::test_containerlab_installed
PASSED                               [  4%]
tests/test_routing_basics.py::TestEnvironment::test_docker_available PASSED
[  9%]
tests/test_routing_basics.py::TestEnvironment::test_ansible_installed PASSED
[ 13%]
tests/test_routing_basics.py::TestEnvironment::test_topology_file_exists
PASSED                                   [ 18%]
tests/test_routing_basics.py::TestTopologyStructure::test_has_correct_name
PASSED                               [ 22%]
tests/test_routing_basics.py::TestTopologyStructure::test_has_six_nodes
PASSED                                    [ 27%]
tests/test_routing_basics.py::TestTopologyStructure::test_has_five_links
PASSED                                  [ 31%]
tests/test_routing_basics.py::TestTopologyStructure::test_no_latest_tags
PASSED                                  [ 36%]
tests/test_routing_basics.py::TestTopologyStructure::test_routers_have_mgmt_
ips PASSED                           [ 40%]
tests/test_routing_basics.py::TestTopologyStructure::test_hosts_have_exec
PASSED                                 [ 45%]
tests/test_routing_basics.py::TestAnsibleStructure::test_inventory_exists
PASSED                                 [ 50%]
tests/test_routing_basics.py::TestAnsibleStructure::test_playbook_exists
PASSED                                   [ 54%]
tests/test_routing_basics.py::TestAnsibleStructure::test_interface_template_
exists PASSED                        [ 59%]
tests/test_routing_basics.py::TestAnsibleStructure::test_routes_template_exi
sts PASSED                           [ 63%]
tests/test_routing_basics.py::TestAnsibleStructure::test_host_vars_exist
PASSED                                   [ 68%]
tests/test_routing_basics.py::TestAnsibleStructure::test_host_vars_have_rout
es PASSED                            [ 72%]
tests/test_routing_basics.py::TestAnsibleStructure::test_inventory_has_three
_routers PASSED                      [ 77%]
```

```
tests/test_routing_basics.py::TestLabDeployment::test_lab_deploys PASSED
[ 81%]
tests/test_routing_basics.py::TestLabDeployment::test_six_containers_running
PASSED                            [ 86%]
tests/test_routing_basics.py::TestConnectivity::test_host1_to_host2 PASSED
[ 90%]
tests/test_routing_basics.py::TestConnectivity::test_host1_to_host3 PASSED
[ 95%]
tests/test_routing_basics.py::TestConnectivity::test_host2_to_host3 PASSED
[100%]

================================ 22 passed in 306.88s (0:05:06)
==============================
```

Completion Checklist

- ✅ Exercise 1: Deployed lab, applied Ansible config, verified end-to-end connectivity
- ✅ Exercise 2: Read routing tables, traced packet path hop by hop
- ✅ Exercise 3: Diagnosed and fixed missing route (return path analysis)
- ✅ Exercise 4: Diagnosed and fixed wrong next-hop (black hole)
- ✅ Exercise 5: Diagnosed and fixed routing loop (TTL)
- ✅ Exercise 6: Diagnosed and fixed unreachable next-hop (link down)

Next Steps

Commit your exercises to your fork and proceed to Lesson 4 (coming soon).